

Linear systems, distances & RANSAC

Predicted quiz questions with answers and the theory behind them

Announced topics: inhomogeneous and homogeneous linear systems of equations, point–line and point–plane distances, RANSAC. For RANSAC, you need to know how many points each minimal model requires (lines, planes, circles, spheres, and so on).

Sources used: *matrix.pdf* (§3, inhomogeneous systems), *Lagrange-1.pdf* (Lagrange multipliers, homogeneous systems, distances, optimal line/plane fitting, SVD), the *Lab 3 RANSAC notebook*, and *CircleEllipseFitting.pdf* (for the circle, sphere and ellipse models).

How to read the question tags

- VERY HIGH** Almost certain to appear, in some variant.
 - HIGH** Likely. It covers a core formula of an announced topic.
 - MEDIUM** Plausible. It uses the same multiple-choice format as Quiz 1, but on a secondary formula.
 - NEW FORMAT** A question type not seen in Quiz 1 but consistent with the topics: true/false, numerical, “which is false”, matrix construction, matching, ordering.
- Every question shows the ✓ **correct answer** together with a short **Why**.

What Quiz 1 tells us about Quiz 2

- Every question asked you to recognize a formula.** The five answers ($\mathbf{a}$, $(\mathbf{A}^\top + \mathbf{A})\mathbf{x}$, $2\mathbf{A}\mathbf{x}$, $(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$, $2\mathbf{x}$) were copied directly from the *matrix.pdf* formulas. There were no calculations and no derivations.
- The wrong answers were near-misses built from the same symbols** ($2\mathbf{x}$, $\mathbf{x}$, $2\mathbf{A}\mathbf{x}$, $\mathbf{A}\mathbf{x}$, $\mathbf{a}$, $\mathbf{A}^\top \mathbf{x}$). The trap is always a small change: a missing transpose, a factor of 2, or the wrong eigenvalue.
- Three of the five questions were variants of one idea** (gradients). Expect two or three variants of “how many points for model X?”.
- One announced topic got no question at all** (circle fitting). The quiz focuses on the headline formulas.
- The pseudo-inverse was already asked.** The inhomogeneous topic will probably come back as the normal equation or “the optimal solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ ”.

Forecast: the five most probable questions

Question	Answer	Tag
Minimal number of points for a line / plane / circle / sphere (2–3 variants)	2 / 3 / 3 / 4 (ellipse: 5)	VERY HIGH
Optimal solution of the homogeneous system $\mathbf{Ax} = \mathbf{0}$	Eigenvector of $\mathbf{A}^T \mathbf{A}$ with the smallest eigenvalue	VERY HIGH
Distance of $\mathbf{x}_0$ from $\mathbf{l}^T \mathbf{x} + m = 0$	$ \mathbf{l}^T \mathbf{x}_0 + m / \sqrt{\mathbf{l}^T \mathbf{l}}$ (or $ \mathbf{l}^T \mathbf{x}_0 + m $ if $\mathbf{l}^T \mathbf{l} = 1$)	HIGH
Optimal solution / normal equation of $\mathbf{Ax} = \mathbf{b}$	$\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$, $\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$	HIGH
Number of RANSAC iterations	$k = \log(1 - p) / \log(1 - w^s)$	MEDIUM

Quiz scoring (lecture 00): 5 questions. 4–5 correct gives +1, 3 correct gives 0, fewer gives −1. The retake has 25 questions and needs 17 correct, so everything here is also useful for the retake.

1 Inhomogeneous overdetermined linear systems

Theory

The problem. We want to solve $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ is $m \times n$, $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{b} \in \mathbb{R}^m$ with $\mathbf{b} \neq \mathbf{0}$. When $m > n$ there are more equations than unknowns (for example, many noisy points), so there is usually *no exact solution*. Instead we minimize the squared difference between the two sides:

$$\mathbf{x}^* = \arg \min_{\mathbf{x}} J(\mathbf{x}), \quad J(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|_2^2 = (\mathbf{Ax} - \mathbf{b})^T (\mathbf{Ax} - \mathbf{b}) = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} - 2\mathbf{x}^T \mathbf{A}^T \mathbf{b} + \mathbf{b}^T \mathbf{b}.$$

Gradient rules (these were the Quiz 1 questions, and you need them here):

$$\frac{\partial \mathbf{a}^T \mathbf{x}}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial \mathbf{x}^T \mathbf{x}}{\partial \mathbf{x}} = 2\mathbf{x}, \quad \frac{\partial \mathbf{x}^T \mathbf{Bx}}{\partial \mathbf{x}} = (\mathbf{B}^T + \mathbf{B})\mathbf{x} \stackrel{\mathbf{B}=\mathbf{B}^T}{=} 2\mathbf{Bx}.$$

$\mathbf{A}^T \mathbf{A}$ is symmetric, so differentiating J and setting the result to zero gives

$$\frac{\partial J}{\partial \mathbf{x}} = 2\mathbf{A}^T \mathbf{Ax} - 2\mathbf{A}^T \mathbf{b} = \mathbf{0}.$$

Normal equation: $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$

Least-squares solution: $\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b} = \mathbf{A}^+ \mathbf{b}$

$\mathbf{A}^+ = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ is the Moore–Penrose pseudo-inverse of a matrix with more rows than columns (and full column rank). If $\mathbf{A}$ is square and invertible, $\mathbf{A}^+ = \mathbf{A}^{-1}$.

Requirements. $\mathbf{A}^T \mathbf{A}$ ($n \times n$) must be invertible. That means $\mathbf{A}$ has full column rank, so you need at least n independent equations ($m \geq n$).

How to recognize an inhomogeneous system, with the course examples

A system is inhomogeneous when one coefficient of the model is fixed (usually to 1). That known term moves to the right-hand side and becomes $\mathbf{b} \neq \mathbf{0}$. This happens with *explicit* models and with the course's circle and sphere formulation.

Model	Row of $\mathbf{A}$ for point i	Entry of $\mathbf{b}$	Unknown vector
Line $y = mx + b$	$[x_i \ 1]$	y_i	$[m \ b]^T$
Plane $z = ax + by + c$	$[x_i \ y_i \ 1]$	z_i	$[a \ b \ c]^T$
Circle $(x - x_0)^2 + (y - y_0)^2 = R^2$	$[1 \ -2x_i \ -2y_i]$	$-x_i^2 - y_i^2$	$[x_0^2 + y_0^2 - R^2 \ x_0 \ y_0]^T$
Sphere	$[1 \ -2x_i \ -2y_i \ -2z_i]$	$-x_i^2 - y_i^2 - z_i^2$	$[x_0^2 + y_0^2 + z_0^2 - R^2 \ x_0 \ y_0 \ z_0]^T$

Closed forms from the handout. For the line, $\begin{bmatrix} m \\ b \end{bmatrix} = \begin{bmatrix} \sum x_i^2 & \sum x_i \\ \sum x_i & N \end{bmatrix}^{-1} \begin{bmatrix} \sum x_i y_i \\ \sum y_i \end{bmatrix}$. For the plane, the matrix is $\begin{bmatrix} \sum x_i^2 & \sum x_i y_i & \sum x_i \\ \sum x_i y_i & \sum y_i^2 & \sum y_i \\ \sum x_i & \sum y_i & N \end{bmatrix}$ and the right-hand side is $[\sum x_i z_i \ \sum y_i z_i \ \sum z_i]^T$. Both are just $\mathbf{A}^T \mathbf{A}$ and $\mathbf{A}^T \mathbf{b}$ written out with sums.

Geometric meaning and limits. Fitting $y = mx + b$ minimizes *vertical* residuals $(y_i - mx_i - b)^2$, not perpendicular distances. The result depends on how the data is rotated, and a vertical line $x = \text{const}$ cannot be represented at all. The same holds for $z = ax + by + c$: it minimizes residuals along z and cannot represent a vertical plane. The implicit form together with a homogeneous system (Sections 2 and 3) fixes both problems.

Typos in the handout (worth knowing in case a question copies them). In matrix.pdf §3, the expansion of J reads “ $-2\mathbf{x}^T \mathbf{A}^T \mathbf{x}$ ”; it should be $-2\mathbf{x}^T \mathbf{A}^T \mathbf{b}$. In §4.1 the $y = mx + b$ system is called “homogeneous”; it is **inhomogeneous**, because the right-hand side $[y_1 \ \dots \ y_N]^T$ is not zero.

Predicted questions

Q1.1 Optimal solution of $\mathbf{Ax} = \mathbf{b}$

HIGH

What is the optimal (least-squares) solution of the overdetermined inhomogeneous linear system $\mathbf{Ax} = \mathbf{b}$?

- $\mathbf{A}^{-1}\mathbf{b}$
- ✓ $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$ **CORRECT**
- $\mathbf{A}^T \mathbf{b}$
- the eigenvector of $\mathbf{A}^T \mathbf{A}$ corresponding to the smallest eigenvalue

Why: Minimizing $\|\mathbf{Ax} - \mathbf{b}\|^2$ gives the normal equation, and solving it gives the pseudo-inverse times $\mathbf{b}$. $\mathbf{A}^{-1}$ does not exist because $\mathbf{A}$ is not square. The eigenvector answer belongs to the *homogeneous* case.

Q1.2 Normal equation

HIGH

What is the normal equation of the overdetermined linear system $\mathbf{Ax} = \mathbf{b}$?

- ✓ $\mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b}$ **CORRECT**
- $\mathbf{AA}^T \mathbf{x} = \mathbf{b}$
- $\mathbf{A}^T \mathbf{Ax} = \lambda \mathbf{x}$
- $\mathbf{Ax} = \mathbf{0}$

Why: Set $\partial J / \partial \mathbf{x} = 2\mathbf{A}^T \mathbf{Ax} - 2\mathbf{A}^T \mathbf{b} = \mathbf{0}$. The eigen-equation $\mathbf{A}^T \mathbf{Ax} = \lambda \mathbf{x}$ is the homogeneous case.

Q1.3 Moore–Penrose pseudo-inverse (repeat from Quiz 1) MEDIUM

What is the Moore–Penrose pseudo-inverse of the matrix $\mathbf{A}$, if $\mathbf{A}$ has more rows than columns?

- ☐ $\mathbf{A}^{-T}$
- ✓ $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ CORRECT
- ☐ $\sqrt{\mathbf{A} \mathbf{A}^T}$
- ☐ $\mathbf{A}^{-1}$

Why: This was Quiz 1, Q4, and it may be reused. Watch for the lookalike $\mathbf{A}^T (\mathbf{A} \mathbf{A}^T)^{-1}$: that is the pseudo-inverse for a *wide* matrix (more columns than rows).

Q1.4 Cost function MEDIUM

Which cost function is minimized when solving the overdetermined inhomogeneous system $\mathbf{A}\mathbf{x} = \mathbf{b}$?

- ☐ $\mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x}$ subject to $\mathbf{x}^T \mathbf{x} = 1$
- ✓ $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2$ CORRECT
- ☐ $\|\mathbf{A}\mathbf{x}\|_2^2$
- ☐ $\|\mathbf{x} - \mathbf{b}\|_2^2$

Why: This is the least-squares (L2) cost. $\|\mathbf{A}\mathbf{x}\|_2^2$ with a unit-length constraint is the homogeneous problem.

Q1.5 Gradient of the least-squares cost MEDIUM

What is the gradient of $J = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ with respect to $\mathbf{x}$?

- ☐ $2\mathbf{A}\mathbf{x} - 2\mathbf{b}$
- ✓ $2\mathbf{A}^T \mathbf{A} \mathbf{x} - 2\mathbf{A}^T \mathbf{b}$ CORRECT
- ☐ $\mathbf{A}^T \mathbf{A} \mathbf{x} - \mathbf{b}$
- ☐ $2\mathbf{A} \mathbf{A}^T \mathbf{x} - 2\mathbf{A} \mathbf{b}$

Why: $J = \mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} - 2\mathbf{x}^T \mathbf{A}^T \mathbf{b} + \mathbf{b}^T \mathbf{b}$. Use $\partial(\mathbf{x}^T \mathbf{B} \mathbf{x})/\partial \mathbf{x} = 2\mathbf{B}\mathbf{x}$ for symmetric $\mathbf{B} = \mathbf{A}^T \mathbf{A}$, and $\partial(\mathbf{x}^T \mathbf{a})/\partial \mathbf{x} = \mathbf{a}$ with $\mathbf{a} = \mathbf{A}^T \mathbf{b}$.

Q1.6 Line fitting in explicit form MEDIUM

Fitting the line $y = mx + b$ to points (x_i, y_i) by least squares, which is the i -th row of the coefficient matrix $\mathbf{A}$ (unknowns $[m \ b]^T$)?

- ☐ $[x_i \ y_i \ 1]$
- ✓ $[x_i \ 1]$, with right-hand side y_i CORRECT
- ☐ $[y_i \ 1]$, with right-hand side x_i
- ☐ $[x_i^2 \ x_i \ 1]$

Why: Each point gives $y_i = m x_i + b \cdot 1$. The system is inhomogeneous because y_i is on the right-hand side.

Q1.7 Plane fitting in explicit form MEDIUM

Fitting the plane $z = ax + by + c$ to 3D points, what is the i -th row of the coefficient matrix, and what type of system is it?

- ☐ $[x_i \ y_i \ z_i \ 1]$, homogeneous
- ✓ $[x_i \ y_i \ 1]$ with right-hand side z_i , inhomogeneous CORRECT
- ☐ $[x_i \ y_i \ z_i]$, inhomogeneous
- ☐ $[1 \ -2x_i \ -2y_i]$, inhomogeneous

Why: $z_i = ax_i + by_i + c$. The row $[x_i \ y_i \ z_i \ 1]$ belongs to the *implicit* (homogeneous) plane $ax + by + cz + d = 0$. The row $[1 \ -2x_i \ -2y_i]$ is circle fitting.

Q1.8 What explicit fitting minimizes MEDIUM

Least-squares fitting of $y = mx + b$ minimizes the sum of squared

- ☐ perpendicular (orthogonal) distances of the points from the line
- ✓ vertical (y -direction) differences between the points and the line CORRECT
- ☐ horizontal (x -direction) differences
- ☐ distances of the points from their center of gravity

Why: The residual is $y_i - (mx_i + b)$, a vertical offset. Perpendicular distances come from implicit line fitting with $a^2 + b^2 = 1$ (Section 3.3).

2 Homogeneous linear systems and Lagrange multipliers

Theory

Lagrange multipliers (Lagrange-1.pdf §1). To minimize $f(\mathbf{x})$ subject to $g(\mathbf{x}) = c$: at the optimum, the level curve of f just touches the constraint curve, so the two **gradients are parallel**, $\nabla f = \lambda \nabla g$. Lagrange put this condition and the constraint into a single cost function:

$$J(\mathbf{x}, \lambda) = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c), \quad \frac{\partial J}{\partial \mathbf{x}} = \mathbf{0} \Rightarrow \nabla f = \lambda \nabla g, \quad \frac{\partial J}{\partial \lambda} = 0 \Rightarrow g(\mathbf{x}) = c.$$

The new scalar λ is the *Lagrange multiplier*. The method works in any dimension, and *setting the derivative with respect to λ to zero gives back the constraint*.

The homogeneous problem (§2). We want to solve $\mathbf{Ax} = \mathbf{0}$ with $m > n$. Two issues arise: (i) $\mathbf{x} = \mathbf{0}$ is always a solution, the *trivial* one; and (ii) if $\mathbf{x}$ is a solution, so is $\alpha \mathbf{x}$ for any α (*scale ambiguity*). Both are removed by fixing the length: $\mathbf{x}^T \mathbf{x} = 1$. Now we minimize $\|\mathbf{Ax}\|^2 = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax}$ subject to $\mathbf{x}^T \mathbf{x} = 1$:

$$J = \mathbf{x}^T \mathbf{A}^T \mathbf{Ax} - \lambda(\mathbf{x}^T \mathbf{x} - 1), \quad \frac{\partial J}{\partial \mathbf{x}} = 2\mathbf{A}^T \mathbf{Ax} - 2\lambda \mathbf{x} = \mathbf{0} \implies \mathbf{A}^T \mathbf{Ax} = \lambda \mathbf{x}.$$

This is a standard **eigenvalue problem**, so the candidate solutions are the eigenvectors of $\mathbf{A}^T \mathbf{A}$. Substituting one back into the cost gives $\mathbf{x}^T \mathbf{A}^T \mathbf{Ax} = \lambda \mathbf{x}^T \mathbf{x} = \lambda$: **the cost equals the eigenvalue**.

Solution of $\min \|\mathbf{Ax}\|$ subject to $\|\mathbf{x}\| = 1$: the eigenvector of $\mathbf{A}^T \mathbf{A}$ corresponding to the **smallest** eigenvalue. The minimal cost is $\lambda_{\min}$.

Maximizing instead gives the eigenvector of the **largest** eigenvalue.

$\mathbf{A}^T \mathbf{A}$ is symmetric and positive semi-definite, so its eigenvalues are **real and non-negative** and its eigenvectors are orthogonal. If $\lambda_{\min} = 0$, an exact (noise-free) solution exists.

Unknowns versus degrees of freedom. A homogeneous system with n unknowns has only $n - 1$ degrees of freedom, because the scale is arbitrary. So it needs at least $n - 1$ independent equations. Examples: the implicit line $ax + by + c = 0$ has 3 unknowns and 2 DoF; the implicit plane $ax + by + cz + d = 0$ has 4 unknowns and 3 DoF; the conic $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$ has 6 unknowns and 5 DoF.

Singular value decomposition (§5). Every $N \times M$ matrix can be written as $\mathbf{A} = \mathbf{USV}^T$, where $\mathbf{U}^T \mathbf{U} = \mathbf{I}$ and $\mathbf{V}^T \mathbf{V} = \mathbf{I}$, and $\mathbf{S}$ is diagonal with $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

- The columns of $\mathbf{V}$ are the eigenvectors of $\mathbf{A}^T \mathbf{A}$. The columns of $\mathbf{U}$ are the eigenvectors of $\mathbf{AA}^T$.
- $\sigma_i = \sqrt{\lambda_i}$. $\mathbf{A}^T \mathbf{A}$ and $\mathbf{AA}^T$ share the same non-zero eigenvalues, and all of them are real and non-negative.
- So the homogeneous solution is the **last column of $\mathbf{V}$** : the right singular vector of the smallest singular value. In numpy: `U, S, Vt = np.linalg.svd(A); x = Vt[-1]`.

Course examples of homogeneous systems.

Model	Row for point i	Unknowns
Implicit line $ax + by + c = 0$	$[x_i \ y_i \ 1]$	$[a \ b \ c]^T$
Implicit plane $ax + by + cz + d = 0$	$[x_i \ y_i \ z_i \ 1]$	$[a \ b \ c \ d]^T$
Ellipse (conic)	$[x_i^2 \ x_i y_i \ y_i^2 \ x_i \ y_i \ 1]$	$[A \ B \ C \ D \ E \ F]^T$

The line and the plane are solved by the smallest eigenvector of $\mathbf{A}^T \mathbf{A}$ (3×3 and 4×4). The ellipse leads to the *generalized* eigenproblem $\mathbf{M}^T \mathbf{M} \mathbf{z} = -\lambda \mathbf{K} \mathbf{z}$. For the ellipse, the constraint $B^2 - 4AC = -1$ (written $\mathbf{z}^T \mathbf{K} \mathbf{z} = -1$) replaces $\mathbf{z}^T \mathbf{z} = 1$ and guarantees an ellipse, not a hyperbola or parabola.

Predicted questions

Q2.1 Optimal solution of $\mathbf{Ax} = \mathbf{0}$

VERY HIGH

What is the optimal (non-trivial) solution of the overdetermined homogeneous linear system $\mathbf{Ax} = \mathbf{0}$?

- the eigenvector of $\mathbf{A}^T\mathbf{A}$ corresponding to the largest eigenvalue
- $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$
- ✓ the eigenvector of $\mathbf{A}^T\mathbf{A}$ corresponding to the smallest eigenvalue CORRECT
- the eigenvector of $\mathbf{AA}^T$ corresponding to the smallest eigenvalue

Why: $\mathbf{A}^T\mathbf{Ax} = \lambda\mathbf{x}$, and the cost $\mathbf{x}^T\mathbf{A}^T\mathbf{Ax} = \lambda$, so the minimum is at the smallest eigenvalue. The traps are *largest* (the maximizer) and $\mathbf{AA}^T$ (the wrong matrix; its eigenvectors are the columns of $\mathbf{U}$).

Q2.2 The constraint

HIGH

Which constraint is applied when the homogeneous system $\mathbf{Ax} = \mathbf{0}$ is solved?

- ✓ $\mathbf{x}^T\mathbf{x} = 1$ CORRECT
- $\mathbf{x} = \mathbf{0}$
- $\mathbf{x}^T\mathbf{Ax} = 0$
- $\mathbf{Ax} = \mathbf{b}$

Why: A unit length rules out the trivial solution $\mathbf{x} = \mathbf{0}$ and fixes the arbitrary scale.

Q2.3 Why a constraint is needed

MEDIUM

Why is a constraint needed for the overdetermined homogeneous problem?

- Because $\mathbf{A}^T\mathbf{A}$ is not invertible otherwise
- ✓ Because $\mathbf{x} = \mathbf{0}$ is a trivial solution, and any scaled solution is also a solution CORRECT
- Because the system has fewer equations than unknowns
- To remove outliers from the data

Why: Without it, the minimum of $\|\mathbf{Ax}\|^2$ is simply $\mathbf{x} = \mathbf{0}$, which is useless.

Q2.4 Lagrangian cost function

MEDIUM

What is the cost function of the homogeneous problem written with a Lagrange multiplier?

- ✓ $\mathbf{x}^T\mathbf{A}^T\mathbf{Ax} - \lambda(\mathbf{x}^T\mathbf{x} - 1)$ CORRECT
- $(\mathbf{Ax} - \mathbf{b})^T(\mathbf{Ax} - \mathbf{b})$
- $\mathbf{x}^T\mathbf{x} - \lambda(\mathbf{x}^T\mathbf{A}^T\mathbf{Ax} - 1)$
- $\mathbf{x}^T\mathbf{A}^T\mathbf{Ax}$

Why: It is the quantity to minimize, $\|\mathbf{Ax}\|^2$, minus λ times (constraint - 1). Option 3 swaps the cost and the constraint. Option 4 has no constraint, so its minimum is $\mathbf{x} = \mathbf{0}$.

Q2.5 Resulting equation

MEDIUM

Setting $\partial J / \partial \mathbf{x} = \mathbf{0}$ for $J = \mathbf{x}^T\mathbf{A}^T\mathbf{Ax} - \lambda(\mathbf{x}^T\mathbf{x} - 1)$ leads to

- $\mathbf{A}^T\mathbf{Ax} = \mathbf{A}^T\mathbf{b}$
- ✓ $\mathbf{A}^T\mathbf{Ax} = \lambda\mathbf{x}$ CORRECT
- $\mathbf{Ax} = \lambda\mathbf{x}$
- $\mathbf{AA}^T\mathbf{x} = \lambda\mathbf{x}$

Why: $2\mathbf{A}^T\mathbf{Ax} - 2\lambda\mathbf{x} = \mathbf{0}$. This is the eigenvalue problem of the symmetric matrix $\mathbf{A}^T\mathbf{A}$.

Q2.6 Value of the cost at the solution

MEDIUM

If $\mathbf{x}$ is a unit eigenvector of $\mathbf{A}^T\mathbf{A}$ with eigenvalue λ , what is $\mathbf{x}^T\mathbf{A}^T\mathbf{Ax}$?

- 1
- 0
- ✓ λ CORRECT
- λ^2

Why: $\mathbf{x}^T\mathbf{A}^T\mathbf{Ax} = \mathbf{x}^T(\lambda\mathbf{x}) = \lambda\mathbf{x}^T\mathbf{x} = \lambda$. That is why the smallest eigenvalue gives the smallest cost.

Q2.7 Maximization

MEDIUM

Which vector *maximizes* $\|\mathbf{Ax}\|^2$ subject to $\mathbf{x}^T \mathbf{x} = 1$?

- the eigenvector of $\mathbf{A}^T \mathbf{A}$ with the smallest eigenvalue
- ✓ the eigenvector of $\mathbf{A}^T \mathbf{A}$ with the largest eigenvalue CORRECT
- $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$
- $\mathbf{x} = \mathbf{0}$

Why: The handout says so explicitly: the maximum corresponds to the largest eigenvalue.

Q2.8 SVD solution

MEDIUM

Using the singular value decomposition $\mathbf{A} = \mathbf{U} \mathbf{S} \mathbf{V}^T$ (singular values in descending order), the solution of the homogeneous system is

- the first column of $\mathbf{U}$
- the last column of $\mathbf{U}$
- the first column of $\mathbf{V}$
- ✓ the last column of $\mathbf{V}$ CORRECT

Why: The columns of $\mathbf{V}$ are eigenvectors of $\mathbf{A}^T \mathbf{A}$, and $\sigma_i = \sqrt{\lambda_i}$ decreases along the diagonal, so the last column goes with the smallest eigenvalue.

Q2.9 Eigenvalues of $\mathbf{A}^T \mathbf{A}$

MEDIUM

The eigenvalues of $\mathbf{A}^T \mathbf{A}$ are always

- complex
- positive and negative in equal numbers
- ✓ real and non-negative CORRECT
- equal to 1

Why: $\mathbf{A}^T \mathbf{A}$ is symmetric (so its eigenvalues are real) and positive semi-definite, since $\mathbf{x}^T \mathbf{A}^T \mathbf{A} \mathbf{x} = \|\mathbf{Ax}\|^2 \geq 0$.

Q2.10 Lagrange multiplier: derivative with respect to λ

MEDIUM

For the Lagrangian $J = f(\mathbf{x}) - \lambda(g(\mathbf{x}) - c)$, what does $\partial J / \partial \lambda = 0$ give?

- $\nabla f = \mathbf{0}$
- ✓ the constraint $g(\mathbf{x}) = c$ CORRECT
- $\lambda = 0$
- $\nabla f = \lambda \nabla g$

Why: $\partial J / \partial \lambda = -(g(\mathbf{x}) - c)$. The parallel-gradient condition $\nabla f = \lambda \nabla g$ comes from $\partial J / \partial \mathbf{x} = \mathbf{0}$ instead.

Q2.11 Lagrange condition at the optimum

MEDIUM

At a constrained optimum of $f(\mathbf{x})$ subject to $g(\mathbf{x}) = c$, the gradients of f and g are

- perpendicular
- ✓ parallel: $\nabla f = \lambda \nabla g$ CORRECT
- both zero
- equal in length

Why: The level curve of f is tangent to the constraint curve, so their normals (gradients) are parallel.

Q2.12 Which formulation is homogeneous?

MEDIUM

Which fitting problem, as formulated in the course, leads to a *homogeneous* linear system?

- line fitting with $y = mx + b$
- plane fitting with $z = ax + by + c$
- circle fitting with unknowns $[x_0^2 + y_0^2 - R^2, x_0, y_0]^T$
- ✓ plane fitting with $ax + by + cz + d = 0$ CORRECT

Why: The implicit plane gives $[x_i \ y_i \ z_i \ 1] [a \ b \ c \ d]^T = 0$. The other three all have a non-zero right-hand side.

3 Point–line and point–plane distances; optimal line/plane fitting

Theory

A **general (hyper)plane** is written as $\mathbf{l}^T \mathbf{x} + m = 0$. For a 2D line, $\mathbf{l} = [a \ b]^T$ and $m = c$ (from $ax + by + c = 0$). For a 3D plane, $\mathbf{l} = [a \ b \ c]^T$ and $m = d$ (from $ax + by + cz + d = 0$). The vector $\mathbf{l}$ is the **normal vector**.

Derivation with a Lagrange multiplier (Lagrange-1.pdf §4.1). Find the point $\mathbf{x}$ on the line/plane that is closest to $\mathbf{x}_0$:

$$J = (\mathbf{x} - \mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) + \lambda (\mathbf{l}^T \mathbf{x} + m), \quad \frac{\partial J}{\partial \mathbf{x}} = 2(\mathbf{x} - \mathbf{x}_0) + \lambda \mathbf{l} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{x}_0 - \frac{\lambda}{2} \mathbf{l}.$$

Substituting into the constraint $\mathbf{l}^T \mathbf{x} + m = 0$ gives $\mathbf{l}^T \mathbf{x}_0 - \frac{\lambda}{2} \mathbf{l}^T \mathbf{l} + m = 0$, so

$$\lambda = 2 \frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}}, \quad \mathbf{x} - \mathbf{x}_0 = -\frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}} \mathbf{l}, \quad d = \|\mathbf{x} - \mathbf{x}_0\| = \frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\mathbf{l}^T \mathbf{l}} \sqrt{\mathbf{l}^T \mathbf{l}} = \frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\sqrt{\mathbf{l}^T \mathbf{l}}}.$$

Distance of $\mathbf{x}_0$ from $\mathbf{l}^T \mathbf{x} + m = 0$: $d = \frac{|\mathbf{l}^T \mathbf{x}_0 + m|}{\sqrt{\mathbf{l}^T \mathbf{l}}}$ If $\mathbf{l}^T \mathbf{l} = 1$: $d = |\mathbf{l}^T \mathbf{x}_0 + m|$ (the implicit equation gives the distance itself).

2D: $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$ **3D:** $d = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$ The formula holds in **any dimension**.

- The **constant term** (c or d) **never goes under the square root**. Only the normal-vector components do.
- Without the absolute value, $\mathbf{l}^T \mathbf{x}_0 + m$ is a *signed* distance: its sign tells you which side of the line/plane $\mathbf{x}_0$ is on.
- The distance of the origin from the line/plane is $|m| / \|\mathbf{l}\|$.
- The closest point (foot of the perpendicular) is $\mathbf{x} = \mathbf{x}_0 - \frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}} \mathbf{l}$.
- The handout writes “ $\mathbf{l}^T \mathbf{x} + m$ ” in the final numerator; it means $\mathbf{l}^T \mathbf{x}_0 + m$, evaluated at the given point.

Worked examples. The point $(1, 2)$ and the line $3x + 4y - 1 = 0$: $d = |3 + 8 - 1| / \sqrt{9 + 16} = 10/5 = 2$. The point $(1, 1, 1)$ and the plane $2x + y + 2z + 4 = 0$: $d = |2 + 1 + 2 + 4| / \sqrt{4 + 1 + 4} = 9/3 = 3$.

Other distances used for RANSAC inlier tests. For a circle or sphere with center $\mathbf{c}$ and radius R : $d = \|\mathbf{p} - \mathbf{c}\| - R$. For a 3D line through $\mathbf{p}_0$ with unit direction $\mathbf{v}$: $d = \|(\mathbf{p} - \mathbf{p}_0) - ((\mathbf{p} - \mathbf{p}_0)^T \mathbf{v}) \mathbf{v}\|$.

Optimal line/plane fitting (Lagrange-1.pdf §4.2–4.3)

This minimizes the sum of squared *perpendicular* distances $\sum_i d_i^2$ with $\mathbf{l}^T \mathbf{l} = 1$. The problem splits into two steps.

Translation. The handout writes the cost as $\sum_i (\mathbf{l}^T (\mathbf{x}_i + \mathbf{t}))^2$. Setting $\partial J / \partial \mathbf{t} = 0$ gives $\sum_i (\mathbf{x}_i + \mathbf{t}) = \mathbf{0}$, so $\mathbf{t} = -\frac{1}{N} \sum_i \mathbf{x}_i$. **The optimal line/plane passes through the center of gravity $\bar{\mathbf{x}}$, which gives $m = -\mathbf{l}^T \bar{\mathbf{x}}$.**

Rotation. Center the points ($\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$) and stack them as the rows of $\mathbf{X}$. The problem becomes the homogeneous system $\mathbf{X} \mathbf{l} = \mathbf{0}$ with $\mathbf{l}^T \mathbf{l} = 1$, so **the normal $\mathbf{l}$ is the eigenvector of $\mathbf{X}^T \mathbf{X}$ with the smallest eigenvalue**. The minimal cost is $\sum d_i^2 = \lambda_{\min}$. For a 2D line, the *direction* is the eigenvector of the *largest* eigenvalue, which is perpendicular to the normal. $\mathbf{X}^T \mathbf{X}$ is the scatter matrix (proportional to the covariance), so this is PCA.

Why it is better than explicit fitting: it uses true geometric distances, gives the same result however the data is rotated, and handles vertical lines and planes. It is also how RANSAC refines its final model from all the inliers.

Predicted questions

Q3.1 General distance formula

HIGH

What is the distance of the point $\mathbf{x}_0$ from the line/plane $\mathbf{l}^T \mathbf{x} + m = 0$?

- ☐ $|\mathbf{l}^T \mathbf{x}_0 + m| / \mathbf{l}^T \mathbf{l}$
- ✓ $|\mathbf{l}^T \mathbf{x}_0 + m| / \sqrt{\mathbf{l}^T \mathbf{l}}$ **CORRECT**
- ☐ $(\mathbf{l}^T \mathbf{x}_0 + m)^2$
- ☐ $\sqrt{\mathbf{l}^T \mathbf{x}_0 + m}$

Why: The distance vector is $-\frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}} \mathbf{l}$, and its length is $|\mathbf{l}^T \mathbf{x}_0 + m| / \|\mathbf{l}\|$. The trap is the missing square root in option 1.

Q3.2 Normalized line/plane

HIGH

What is the distance of $\mathbf{x}_0$ from $\mathbf{l}^T \mathbf{x} + m = 0$ if $\mathbf{l}^T \mathbf{l} = 1$?

- ✓ $|\mathbf{l}^T \mathbf{x}_0 + m|$ **CORRECT**
- ☐ $(\mathbf{l}^T \mathbf{x}_0 + m)^2$
- ☐ $|\mathbf{l}^T \mathbf{x}_0 + m| / m$
- ☐ $\mathbf{l}^T \mathbf{x}_0$

Why: The denominator is 1, so the implicit equation evaluated at the point is the distance. This is why the Lab 3 notebook uses `abs(a*x+b*y+c)`.

Q3.3 Point–line distance in 2D

HIGH

What is the distance of the point $[x_0 \ y_0]^T$ from the line $ax + by + c = 0$?

- ☐ $|ax_0 + by_0 + c| / \sqrt{a^2 + b^2 + c^2}$
- ✓ $|ax_0 + by_0 + c| / \sqrt{a^2 + b^2}$ **CORRECT**
- ☐ $|ax_0 + by_0 + c| / (a^2 + b^2)$
- ☐ $|ax_0 + by_0|$

Why: Here $\mathbf{l} = [a \ b]^T$ and $m = c$, so c is not under the root.

Q3.4 Point–plane distance in 3D

HIGH

What is the distance of the point $[x_0 \ y_0 \ z_0]^T$ from the plane $ax + by + cz + d = 0$?

- ☐ $|ax_0 + by_0 + cz_0 + d| / \sqrt{a^2 + b^2 + c^2 + d^2}$
- ✓ $|ax_0 + by_0 + cz_0 + d| / \sqrt{a^2 + b^2 + c^2}$ **CORRECT**
- ☐ $|ax_0 + by_0 + cz_0 + d| / (a^2 + b^2 + c^2)$
- ☐ $|ax_0 + by_0 + cz_0|$

Why: Here $\mathbf{l} = [a \ b \ c]^T$ and $m = d$.

Q3.5 The distance vector

MEDIUM

What is the vector from $\mathbf{x}_0$ to its closest point $\mathbf{x}$ on $\mathbf{l}^T \mathbf{x} + m = 0$?

- ✓ $\mathbf{x} - \mathbf{x}_0 = -\frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}} \mathbf{l}$ **CORRECT**
- ☐ $\mathbf{x} - \mathbf{x}_0 = \mathbf{l}$
- ☐ $\mathbf{x} - \mathbf{x}_0 = -(\mathbf{l}^T \mathbf{x}_0 + m) \mathbf{x}_0$
- ☐ $\mathbf{x} - \mathbf{x}_0 = \frac{\mathbf{l}}{\mathbf{l}^T \mathbf{l}}$

Why: The vector is parallel to the normal $\mathbf{l}$, and its length is the distance. It follows from $\mathbf{x} = \mathbf{x}_0 - \frac{\lambda}{2} \mathbf{l}$ with $\lambda = 2 \frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}}$.

Q3.6 Dimension of validity

MEDIUM

For which dimensions is the formula $d = |\mathbf{l}^T \mathbf{x}_0 + m| / \sqrt{\mathbf{l}^T \mathbf{l}}$ valid?

- ☐ only 2D (lines)
- ☐ only 3D (planes)
- ☐ only 2D and 3D
- ✓ arbitrary dimension (hyperplanes) **CORRECT**

Why: The derivation never uses the dimension. The handout says: “this distance calculation is valid for arbitrary dimension”.

Q3.7 Optimal translation**MEDIUM**

In optimal line/plane fitting (minimizing squared perpendicular distances), which point does the fitted line/plane pass through?

- the origin
- ✓ **the center of gravity of the points,**
 $\frac{1}{N} \sum_i \mathbf{x}_i$ **CORRECT**
- the first data point
- the point farthest from the origin

Why: $\partial J / \partial \mathbf{t} = 0$ gives $\mathbf{t} = -\sum \mathbf{x}_i / N$.

Q3.8 Normal of the optimal line/plane**MEDIUM**

After moving the origin to the center of gravity, the normal vector of the optimal line/plane is

- the eigenvector of $\mathbf{X}^T \mathbf{X}$ with the largest eigenvalue
- ✓ **the eigenvector of $\mathbf{X}^T \mathbf{X}$ with the smallest eigenvalue** **CORRECT**
- the center of gravity itself
- $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{1}$

Why: It is the homogeneous problem $\mathbf{X} \mathbf{l} = \mathbf{0}$ with $\mathbf{l}^T \mathbf{l} = 1$. The largest eigenvector gives the *direction* of a 2D line, not its normal.

4 RANSAC: robust fitting

Theory

Why robust fitting? Least squares *squares* the residuals, so a single far-away outlier can dominate the cost and pull the solution off. **RANSAC** (*R*ANdom *S*Ample *C*onsensus) instead looks for the model that the largest set of points agrees with (the *consensus set*), and ignores the rest.

The algorithm (Lab 3 notebook).

1. **Random sampling:** pick a *minimal* random subset of s points (2 for a line).
2. **Model fitting:** compute the model from these points only.
3. **Distance calculation:** compute the distance of *every* point to the model.
4. **Inlier counting:** points with distance $<$ threshold are inliers ($\sigma = 10$ px in the notebook).
5. **Best model update:** if there are more inliers than for the best model so far, store this model and its inliers.
6. **Adaptive iteration:** re-estimate the inlier ratio w from the best inlier count and update the number of iterations k .
7. **After the loop:** refit the model by least squares on *all inliers* of the best model (for a line or plane: center of gravity plus smallest eigenvector).

Line through two points (notebook fit_line_two_points). Let $\Delta x = x_2 - x_1$ and $\Delta y = y_2 - y_1$. The direction is $(\Delta x, \Delta y)$, so a normal is $(-\Delta y, \Delta x)$. Normalize it to get $(a, b) = (-\Delta y, \Delta x) / \sqrt{\Delta x^2 + \Delta y^2}$, then set $c = -(ax_1 + by_1)$. Because $a^2 + b^2 = 1$, the distance of a point is simply $|ax + by + c|$. If the two points coincide (norm $< 10^{-8}$) the sample is *degenerate* and is skipped.

Number of iterations. Let w be the inlier ratio, s the minimal sample size and p the desired confidence.

- The probability that a sample contains only inliers is w^s .
- The probability that it contains at least one outlier is $1 - w^s$.
- The probability that all k samples contain an outlier is $(1 - w^s)^k$.

Requiring $(1 - w^s)^k \leq 1 - p$ and taking logarithms gives:

$$k = \left\lceil \frac{\log(1-p)}{\log(1-w^s)} \right\rceil \quad (\text{the logarithm base does not matter, because it cancels})$$

$p = 0.99$	$s = 2$ (line)	$s = 3$ (plane, circle)	$s = 4$ (sphere)	$s = 5$ (ellipse)
$w = 0.8$	5	7	9	12
$w = 0.7$	7	11	17	26
$w = 0.5$	17	35	72	146
$w = 0.3$	49	169	567	1893

- k does **not** depend on the total number of points N (only on the ratio w).
- k grows **exponentially with** s . This is why RANSAC uses *minimal* samples.
- k grows when w decreases (more outliers) and when p increases.
- **Adaptive RANSAC**: w is not known in advance. Start from a maximum (2000 in the notebook) and recompute k whenever a better model is found. In the notebook ($p = 0.95$, $s = 2$), k fell $256 \rightarrow 125 \rightarrow 10$, ending with 517/1000 inliers ($w \approx 0.52$).
- RANSAC is **non-deterministic**: different runs can return slightly different models.
- The threshold should match the noise level of the inliers.
- **Multi-model fitting** (sequential RANSAC): find one model, remove its inliers, and run again on the remaining points.

Minimal sample sizes

Rule: $s = \left\lceil \frac{\text{degrees of freedom of the model}}{\text{constraints given by one point}} \right\rceil$. A point on a curve in 2D, or on a surface in 3D, gives **one** equation. A point on a curve in 3D (a 3D line or 3D circle) gives **two**.

Model	Parameters	DoF	Min. points	Degenerate sample
2D line	a, b, c up to scale	2	2	coincident points
Plane	a, b, c, d up to scale	3	3	collinear points
2D circle	x_0, y_0, R	3	3	collinear points
Sphere	x_0, y_0, z_0, R	4	4	coplanar points
Ellipse / conic	$A, \dots, F$ up to scale	5	5	e.g. 4 collinear points
Parabola $y = ax^2 + bx + c$	a, b, c	3	3	two points with equal x
3D line	point + direction	4	2 (2 eq./pt)	coincident points
3D circle	center, normal, R	6	3 (2 eq./pt)	collinear points
Circle, known R	x_0, y_0	2	2 (2 sol.)	coincident points
Sphere, known R	x_0, y_0, z_0	3	3 (2 sol.)	collinear points
Cylinder	axis (4) + radius	5	5 (2 w/ normals)	—

Link to the linear systems: the circle's inhomogeneous system has 3 unknowns, so it needs 3 equations (3 points). The implicit plane has 4 unknowns up to scale, so 3 DoF and 3 points. The conic has 6 unknowns up to scale, so 5 points.

Predicted questions: minimal samples (expect 2–3 of these)**Q4.1 2D line****VERY HIGH**

How many points are required to estimate a 2D line minimally?

- ☐ 1
- ✓ **2 CORRECT**
- ☐ 3
- ☐ 4

Why: 2 DoF, and each point gives one equation.

Q4.5 Ellipse**HIGH**

How many points are required to estimate an ellipse (general conic) minimally?

- ☐ 3
- ☐ 4
- ✓ **5 CORRECT**
- ☐ 6

Why: 6 coefficients $A, \dots, F$ up to scale gives 5 DoF. The trap is 6.

Q4.2 Plane**VERY HIGH**

How many points are required to estimate a plane minimally?

- ☐ 2
- ✓ **3 CORRECT**
- ☐ 4
- ☐ 5

Why: 4 coefficients up to scale gives 3 DoF. The points must not be collinear. The trap is 4 (counting coefficients instead of DoF).

Q4.6 3D line**MEDIUM**

How many points are required to estimate a line in 3D space minimally?

- ☐ 1
- ✓ **2 CORRECT**
- ☐ 3
- ☐ 4

Why: A 3D line has 4 DoF, but each point on it gives 2 constraints. In any dimension, two points determine a line.

Q4.3 2D circle**VERY HIGH**

How many points are required to estimate a 2D circle minimally?

- ☐ 2
- ✓ **3 CORRECT**
- ☐ 4
- ☐ 5

Why: The unknowns are x_0, y_0, R . The points must not be collinear.

Q4.7 Parabola**MEDIUM**

How many points are required to estimate the parabola $y = ax^2 + bx + c$ minimally?

- ☐ 2
- ✓ **3 CORRECT**
- ☐ 4
- ☐ 5

Why: 3 unknowns and one linear (inhomogeneous) equation per point.

Q4.4 Sphere**VERY HIGH**

How many points are required to estimate a sphere minimally?

- ☐ 2
- ☐ 3
- ✓ **4 CORRECT**
- ☐ 5

Why: The unknowns are x_0, y_0, z_0, R . The points must not be coplanar. 3 is the common mistake: three points only fix a circle on the sphere.

Predicted questions: the RANSAC algorithm

Q4.8 Number of iterations

MEDIUM

What is the required number of RANSAC iterations (p : confidence, w : inlier ratio, s : sample size)?

- ☐ $\log(p) / \log(w^s)$
- ✓ $\log(1 - p) / \log(1 - w^s)$ **CORRECT**
- ☐ $\log(1 - w) / \log(1 - p^s)$
- ☐ $\log(1 - p) / \log(1 - w)^s$

Why: Require $(1 - w^s)^k \leq 1 - p$, then take logs. Remember: "1 minus" appears in both places, and only w is raised to the power s .

Q4.9 All-inlier probability

MEDIUM

What is the probability that a random sample of s points contains only inliers (w : inlier ratio)?

- ✓ w^s **CORRECT**
- ☐ $1 - w^s$
- ☐ $(1 - w)^s$
- ☐ $s \cdot w$

Why: There are s independent draws, each an inlier with probability w . $1 - w^s$ is the probability that the sample is contaminated, and $(1 - w)^s$ is the probability that *all* points are outliers.

Q4.10 The name

MEDIUM

What does RANSAC stand for?

- ☐ Random Sampling and Classification
- ✓ **Random Sample Consensus** **CORRECT**
- ☐ Robust Adaptive Numerical Sample Correction
- ☐ Randomized Nonlinear Least Squares

Why: RANDOM Sample Consensus. The consensus set is the set of inliers that agree with the model.

Q4.11 What the iteration count does not depend on

MEDIUM

Which quantity does *not* directly influence the required number of RANSAC iterations?

- ☐ the inlier ratio
- ☐ the size of the minimal sample
- ☐ the required confidence
- ✓ **the total number of points** **CORRECT**

Why: The formula contains only p , w and s . N matters only through $w = \text{inliers} / N$.

Q4.12 Why minimal samples?

MEDIUM

Why does RANSAC estimate the model from a *minimal* sample?

- ☐ because least squares cannot handle more points
- ✓ **because the probability of an outlier-free sample, w^s , decreases exponentially with s** **CORRECT**
- ☐ because it gives the most accurate final model
- ☐ because it removes the noise of the inliers

Why: A smaller s gives more all-inlier samples, so fewer iterations are needed. Accuracy comes later, from refitting on all inliers.

Q4.13 Selecting the best model

MEDIUM

In standard RANSAC, which model is selected as the best?

- ☐ the one with the smallest sum of squared distances over all points
- ✓ **the one with the largest number of inliers** **CORRECT**
- ☐ the model of the first iteration
- ☐ the average of all sampled models

Why: The largest consensus set wins. Inliers are the points whose distance is below the threshold.

Q4.14 Adaptive update**MEDIUM**

In adaptive RANSAC, what happens to the planned number of iterations when a model with more inliers is found?

- it increases
- ✓ **it decreases** **CORRECT**
- it stays the same
- it is reset to the maximum

Why: A higher estimated w makes w^s larger and k smaller ($256 \rightarrow 125 \rightarrow 10$ in the notebook).

Q4.15 After the loop**MEDIUM**

What is usually done after the RANSAC iterations finish?

- ✓ **the model is refit with least squares using all inliers of the best model** **CORRECT**
- the minimal sample of the best model is returned unchanged
- the threshold is doubled and RANSAC is rerun
- the outliers are fitted with a second model automatically

Why: A model built from a minimal sample is noisy. Least squares on all inliers gives the accurate final estimate.

Q4.16 Inlier criterion**MEDIUM**

When is a point counted as an inlier of the current RANSAC model?

- if it was part of the minimal sample
- ✓ **if its distance from the model is below a threshold** **CORRECT**
- if it is closer to the center of gravity than the threshold
- if its residual is negative

Why: For a normalized line this is $|ax + by + c| < \sigma$. For a circle or sphere it is $|\|\mathbf{p} - \mathbf{c}\| - R| < \sigma$.

5 New question formats: be ready for a change of style

Quiz 1 only asked “pick the formula” questions, but the lecturer could vary the format. Each item below is consistent with the announced topics and can be answered from the theory in Sections 1–4.

5.1 True or false

Q5.1 True/false statements

NEW FORMAT

Statement	Answer
(a) The number of RANSAC iterations increases if the total number of points increases (with the same inlier ratio).	False
(b) For $\mathbf{Ax} = \mathbf{0}$ with $\mathbf{x}^T \mathbf{x} = 1$, the eigenvector of $\mathbf{A}^T \mathbf{A}$ with the largest eigenvalue minimizes $\ \mathbf{Ax}\ $.	False
(c) The eigenvalues of $\mathbf{A}^T \mathbf{A}$ are always real and non-negative.	True
(d) Least-squares fitting of $y = mx + b$ minimizes the perpendicular distances of the points from the line.	False
(e) The optimal plane in the perpendicular-distance sense passes through the center of gravity of the points.	True
(f) If $\mathbf{I}^T \mathbf{I} = 1$, then $\mathbf{I}^T \mathbf{x}_0 + m$ is the signed distance of $\mathbf{x}_0$ from the line/plane $\mathbf{I}^T \mathbf{x} + m = 0$.	True
(g) In general, a sphere can be estimated from 3 points.	False
(h) RANSAC always returns exactly the same model on the same data.	False
(i) If $\mathbf{A}$ is square and invertible, then $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T = \mathbf{A}^{-1}$.	True
(j) Circle fitting as presented in the course leads to a homogeneous linear system.	False
(k) Using larger-than-minimal samples in RANSAC reduces the required number of iterations.	False
(l) The Lagrangian's derivative with respect to λ gives back the constraint.	True
(m) The point-plane distance formula contains the constant term d under the square root.	False

Why: (a) k depends only on p, w, s . (b) The largest eigenvalue is the maximizer. (d) It minimizes vertical residuals. (g) A sphere needs 4 points. (h) RANSAC is random. (i) $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T = \mathbf{A}^{-1} \mathbf{A}^{-T} \mathbf{A}^T = \mathbf{A}^{-1}$. (j) The circle system is inhomogeneous: its right-hand side is $-x_i^2 - y_i^2$. (k) A larger s makes w^s smaller, so more iterations are needed. (m) Only $\sqrt{a^2 + b^2 + c^2}$ appears in the denominator.

5.2 “Which statement is FALSE?”

Q5.2 Homogeneous systems

NEW FORMAT

Which statement about the overdetermined homogeneous system $\mathbf{Ax} = \mathbf{0}$ is **false**?

- ☐ $\mathbf{x} = \mathbf{0}$ is always a solution
- ☐ the solution is only determined up to scale
- ✓ ☒ the solution is $\mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$ **CORRECT**
- ☐ the solution can be obtained from the SVD of $\mathbf{A}$

Why: That formula is the inhomogeneous solution. Here $\mathbf{b} = \mathbf{0}$, so it would give only the trivial $\mathbf{x} = \mathbf{0}$.

Q5.3 RANSAC

NEW FORMAT

Which statement about RANSAC is **false**?

- ☐ it can fit a model even if about half of the points are outliers
- ☐ the number of iterations grows as the inlier ratio decreases
- ✓ ☒ it needs to know the exact number of outliers in advance **CORRECT**
- ☐ inliers are selected using a distance threshold

Why: The inlier ratio is estimated adaptively from the best model found so far. The notebook fits a line with 50% outliers.

Q5.4 Minimal samples

NEW FORMAT

Which model / minimal-sample pair is **wrong**?

- ☐ 2D line: 2 points
- ☐ plane: 3 points
- ✓ ☒ sphere: 3 points **CORRECT**
- ☐ ellipse: 5 points

Why: A sphere has 4 parameters, so it needs 4 non-coplanar points.

5.3 Numerical questions

Q5.5 Point–line distance

NEW FORMAT

What is the distance of the point (1,2) from the line $3x + 4y - 1 = 0$?

- ☐ 10
- ✓ ☒ 2 **CORRECT**
- ☐ $10/\sqrt{26}$
- ☐ 1/5

Why: $|3 \cdot 1 + 4 \cdot 2 - 1| / \sqrt{3^2 + 4^2} = 10/5 = 2$. The value $10/\sqrt{26}$ wrongly puts c under the root.

Q5.6 Point–plane distance

NEW FORMAT

What is the distance of the point (1,1,1) from the plane $2x + y + 2z + 4 = 0$?

- ☐ 1
- ☐ 9
- ✓ ☒ 3 **CORRECT**
- ☐ $9/\sqrt{13}$

Why: $|2 + 1 + 2 + 4| / \sqrt{4 + 1 + 4} = 9/3 = 3$.

Q5.7 Distance from the origin**NEW FORMAT**

What is the distance of the origin from the plane $2x - y + 2z - 6 = 0$?

- ☐ 6
- ✓ **2** **CORRECT**
- ☐ $6/\sqrt{13}$
- ☐ 3

Why: $|m|/\|1\| = 6/\sqrt{4+1+4} = 6/3 = 2$.

Q5.8 RANSAC iterations**NEW FORMAT**

Plane fitting with RANSAC, 50% inliers, confidence 0.99. Approximately how many iterations are required?

- ☐ 17
- ✓ **35** **CORRECT**
- ☐ 72
- ☐ 146

Why: A plane needs $s = 3$, so $\ln(0.01)/\ln(1 - 0.125) = -4.605/-0.1335 \approx 34.5$, which rounds up to 35. (17 is for a line, 72 for a sphere, 146 for an ellipse.)

Q5.9 All-inlier sample**NEW FORMAT**

With 50% inliers, what is the probability that a random sample for circle fitting contains only inliers?

- ☐ 0.5
- ☐ 0.25
- ✓ **0.125** **CORRECT**
- ☐ 0.875

Why: A circle needs $s = 3$, and $0.5^3 = 0.125$. The value $0.875 = 1 - w^s$ is the probability that the sample is contaminated.

Q5.10 Line from two points (as in the notebook)**NEW FORMAT**

The normalized line through $(1, 1)$ and $(4, 5)$ is written as $ax + by + c = 0$ with $a^2 + b^2 = 1$ and $(a, b) = (-\Delta y, \Delta x)/\|\cdot\|$. What is it?

- ☐ $0.6x + 0.8y - 1.4 = 0$
- ✓ **$-0.8x + 0.6y + 0.2 = 0$** **CORRECT**
- ☐ $-4x + 3y + 1 = 0$
- ☐ $0.8x + 0.6y - 1.4 = 0$

Why: $\Delta = (3, 4)$, $\|\Delta\| = 5$, so $(a, b) = (-0.8, 0.6)$, and $c = -(-0.8 \cdot 1 + 0.6 \cdot 1) = 0.2$. Check with $(4, 5)$: $-3.2 + 3 + 0.2 = 0$. Option 3 is the same line but not normalized.

Q5.11 Explicit line fit**NEW FORMAT**

Fit $y = mx + b$ by least squares to the points $(0, 1)$, $(1, 3)$, $(2, 5)$. What are m and b ?

- ☐ $m = 1, b = 2$
- ✓ **$m = 2, b = 1$** **CORRECT**
- ☐ $m = 3, b = 0$
- ☐ $m = 2, b = 0$

Why: $\begin{bmatrix} \sum x^2 & \sum x \\ \sum x & N \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 3 \end{bmatrix}$ and the right-hand side is $\begin{bmatrix} \sum xy \\ \sum y \end{bmatrix} = \begin{bmatrix} 13 \\ 9 \end{bmatrix}$. Solving gives $m = 2, b = 1$. The points lie exactly on $y = 2x + 1$.

Q5.12 Eigenvalues and the cost**NEW FORMAT**

Suppose $\mathbf{A}^T \mathbf{A}$ has eigenvalues 9, 4 and 0.5 with unit eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. Which vector solves $\min \|\mathbf{Ax}\|^2$ subject to $\|\mathbf{x}\| = 1$, and what is the minimal cost?

- ☐ $\mathbf{v}_1$, cost 9
- ☐ $\mathbf{v}_3$, cost $\sqrt{0.5}$
- ✓ **$\mathbf{v}_3$, cost 0.5** **CORRECT**
- ☐ $\mathbf{v}_2$, cost 4

Why: The smallest eigenvalue wins, and the cost equals the eigenvalue itself. $\sqrt{0.5}$ would be the singular value, which is $\|\mathbf{Ax}\|$ rather than $\|\mathbf{Ax}\|^2$.

Q5.13 Center of gravity**NEW FORMAT**

The optimal line (perpendicular distances) for the points $(0,0)$, $(2,0)$, $(1,3)$ must pass through

- ☐ $(0,0)$
- ✓ ☒ $(1,1)$ **CORRECT**
- ☐ $(1,1.5)$
- ☐ $(1,3)$

Why: The center of gravity is $(\frac{0+2+1}{3}, \frac{0+0+3}{3}) = (1,1)$.

5.4 Matrix construction**Q5.14 Sphere fitting row****NEW FORMAT**

In the linear sphere-fitting system with unknowns $[x_0^2 + y_0^2 + z_0^2 - R^2, x_0, y_0, z_0]^T$, which row and right-hand side belong to the point (x_i, y_i, z_i) ?

- ☐ $[x_i \ y_i \ z_i \ 1]$, right-hand side 0
- ✓ ☒ $[1 \ -2x_i \ -2y_i \ -2z_i]$, right-hand side $-x_i^2 - y_i^2 - z_i^2$ **CORRECT**
- ☐ $[1 \ 2x_i \ 2y_i \ 2z_i]$, right-hand side $x_i^2 + y_i^2 + z_i^2$
- ☐ $[x_i^2 \ y_i^2 \ z_i^2 \ 1]$, right-hand side R^2

Why: Expanding $(x - x_0)^2 + \dots = R^2$ gives $x^2 - 2x_0x + x_0^2 + \dots - R^2 = 0$. Move the squared data terms to the right-hand side. The system is inhomogeneous, with 4 unknowns (hence 4 points).

Q5.15 Ellipse fitting row**NEW FORMAT**

In homogeneous ellipse fitting $\mathbf{Mz} = \mathbf{0}$ with $\mathbf{z} = [A \ B \ C \ D \ E \ F]^T$, which is the row for the point (x_i, y_i) ?

- ☐ $[x_i \ y_i \ 1]$
- ☐ $[1 \ -2x_i \ -2y_i]$
- ✓ ☒ $[x_i^2 \ x_i y_i \ y_i^2 \ x_i \ y_i \ 1]$ **CORRECT**
- ☐ $[x_i^2 \ y_i^2 \ 1]$

Why: It follows from $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$. There are 6 unknowns up to scale, so 5 points are needed.

Q5.16 Implicit plane fitting matrix**NEW FORMAT**

Implicit plane fitting

$[x_i \ y_i \ z_i \ 1] [a \ b \ c \ d]^T = 0$: the solution is the smallest eigenvector of which matrix?

- ☐ $[\sum x_i \ \sum y_i \ \sum z_i \ N]$
- ✓ ☒ $\mathbf{A}^T \mathbf{A} = \sum_i [x_i \ y_i \ z_i \ 1]^T [x_i \ y_i \ z_i \ 1]$, a 4×4 matrix **CORRECT**
- ☐ $\mathbf{A} \mathbf{A}^T$, an $N \times N$ matrix
- ☐ the 3×3 identity

Why: Its entries are sums such as $\sum x_i^2, \sum x_i y_i, \dots, N$ (see the handout, §3). $\mathbf{A} \mathbf{A}^T$ is $N \times N$, and its eigenvectors are the columns of $\mathbf{U}$, not the solution.

5.5 Classification and matching

Q5.17 Homogeneous or inhomogeneous?

NEW FORMAT

Classify each formulation.

Formulation	Type
Line $y = mx + b$	inhomogeneous
Line $ax + by + c = 0$	homogeneous
Plane $z = ax + by + c$	inhomogeneous
Plane $ax + by + cz + d = 0$	homogeneous
Circle, unknowns $[x_0^2 + y_0^2 - R^2, x_0, y_0]$	inhomogeneous
Sphere, unknowns $[x_0^2 + y_0^2 + z_0^2 - R^2, x_0, y_0, z_0]$	inhomogeneous
Ellipse $Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$	homogeneous (with constraint)

Why: Inhomogeneous means there is a known non-zero right-hand side (one coefficient fixed to 1). Homogeneous means everything is multiplied by the unknowns and the right-hand side is 0.

Q5.18 Match each model with its minimal sample

NEW FORMAT

2D line	plane	2D circle	sphere	ellipse	3D line
2	3	3	4	5	2

Why: $s = \lceil \text{DoF} / \text{constraints per point} \rceil$.

5.6 Ordering and short answers

Q5.19 Put the RANSAC steps in order

NEW FORMAT

The steps are: (A) count inliers with a threshold; (B) refit with least squares on all inliers; (C) select a random minimal sample; (D) update the best model and the iteration number; (E) fit the model to the sample.

Answer: $C \rightarrow E \rightarrow A \rightarrow D$, repeated k times, then B.

Q5.20 Short answers

NEW FORMAT

(a) What is the Lagrange multiplier for the point–line distance problem?

Answer: $\lambda = 2 \frac{\mathbf{l}^T \mathbf{x}_0 + m}{\mathbf{l}^T \mathbf{l}}.$

(b) What does $\lambda_{\min}$ of the centered scatter matrix $\mathbf{X}^T \mathbf{X}$ represent in optimal plane fitting?

Answer: The minimal sum of squared point–plane distances.

(c) How are the singular values of $\mathbf{A}$ related to the eigenvalues of $\mathbf{A}^T \mathbf{A}$?

Answer: $\sigma_i = \sqrt{\lambda_i}.$

(d) What is the distance of a point $\mathbf{p}$ from a circle or sphere with center $\mathbf{c}$ and radius R ?

Answer: $|\|\mathbf{p} - \mathbf{c}\| - R|.$

(e) What happens to the number of RANSAC iterations when the required confidence p increases?

Answer: It increases, because $\log(1 - p)$ becomes more negative.

(f) Why is it unnecessary to divide by $\sqrt{a^2 + b^2}$ in the Lab 3 distance function?

Answer: The line is normalized so that $a^2 + b^2 = 1.$

Last-minute cheat sheet

Inhomogeneous: $\mathbf{Ax} = \mathbf{b}$, with $\mathbf{b} \neq \mathbf{0}$ and $m > n$

Cost $\|\mathbf{Ax} - \mathbf{b}\|^2 \rightarrow$ gradient $2\mathbf{A}^\top \mathbf{Ax} - 2\mathbf{A}^\top \mathbf{b} \rightarrow$ normal equation $\mathbf{A}^\top \mathbf{Ax} = \mathbf{A}^\top \mathbf{b} \rightarrow \mathbf{x} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{b} = \mathbf{A}^+ \mathbf{b}$.

Examples: $y = mx + b$ (row $[x_i \ 1]$), $z = ax + by + c$ (row $[x_i \ y_i \ 1]$), circle (row $[1 \ -2x_i \ -2y_i]$), sphere (row $[1 \ -2x_i \ -2y_i \ -2z_i]$).

Gradients: $\partial \mathbf{a}^\top \mathbf{x} = \mathbf{a}$, $\partial \mathbf{x}^\top \mathbf{x} = 2\mathbf{x}$, $\partial \mathbf{x}^\top \mathbf{Ax} = (\mathbf{A}^\top + \mathbf{A})\mathbf{x}$, which is $2\mathbf{Ax}$ if $\mathbf{A}$ is symmetric.

Homogeneous: $\mathbf{Ax} = \mathbf{0}$

Constraint $\mathbf{x}^\top \mathbf{x} = 1$ (avoids $\mathbf{x} = \mathbf{0}$ and fixes the scale) $\rightarrow J = \mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} - \lambda(\mathbf{x}^\top \mathbf{x} - 1) \rightarrow \mathbf{A}^\top \mathbf{Ax} = \lambda \mathbf{x}$.

Cost $= \lambda$. **Minimum: eigenvector of $\mathbf{A}^\top \mathbf{A}$ with the SMALLEST eigenvalue** (maximum: the largest).

SVD $\mathbf{A} = \mathbf{USV}^\top$: solution = **last column of $\mathbf{V}$** , and $\sigma_i = \sqrt{\lambda_i}$. Eigenvalues of $\mathbf{A}^\top \mathbf{A}$ are real and ≥ 0 .

Lagrange: $\partial J / \partial \mathbf{x} = \mathbf{0}$ gives $\nabla f = \lambda \nabla g$ (parallel gradients), and $\partial J / \partial \lambda = 0$ gives back the constraint.

Distances

$d = \frac{|\mathbf{1}^\top \mathbf{x}_0 + m|}{\sqrt{\mathbf{1}^\top \mathbf{1}}}$ (if $\mathbf{1}^\top \mathbf{1} = 1$: $d = |\mathbf{1}^\top \mathbf{x}_0 + m|$), valid in any dimension.

2D: $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$ 3D: $\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$ circle/sphere: $|\|\mathbf{p} - \mathbf{c}\| - R|$.

Optimal line/plane: it passes through the **center of gravity**, and its normal is the **smallest eigenvector** of the centered $\mathbf{X}^\top \mathbf{X}$.

RANSAC

Loop: minimal random sample $\rightarrow$ fit $\rightarrow$ distances $\rightarrow$ inliers (below the threshold) $\rightarrow$ keep the model with the most inliers $\rightarrow$ update k . At the end, refit with least squares on all inliers.

$k = \left\lceil \frac{\log(1-p)}{\log(1-w^s)} \right\rceil$. The all-inlier probability is w^s . k is independent of N and grows exponentially with s . For $p = 0.99$, $w = 0.5$: $s = 2 \rightarrow 17$, $3 \rightarrow 35$, $4 \rightarrow 72$, $5 \rightarrow 146$.

2D line	plane	circle	sphere	ellipse	parabola	3D line	3D circle
2	3	3	4	5	3	2	3

Near-misses to watch for: smallest vs. largest eigenvalue · $\mathbf{A}^\top \mathbf{A}$ vs. $\mathbf{AA}^\top$ · $\mathbf{V}$ vs. $\mathbf{U}$ · $\sqrt{\mathbf{1}^\top \mathbf{1}}$ vs. $\mathbf{1}^\top \mathbf{1}$ · the constant term is never under the root · $\log(1-p) / \log(1-w^s)$ vs. $\log p / \log w^s$ · w^s vs. $1 - w^s$ vs. $(1-w)^s$ · sphere = 4 (not 3) · ellipse = 5 (not 6) · plane = 3 (not 4).